

Program 4: Integrate Git with Docker for Source-Controlled Application Builds

Project Structure

DEVOPSLAB

```
|--Program4
    |--Dockerfile
    |--index.js
    |--package.json
    |--package-lock.json
```

Step 1: Execute the following command to generate the package.json file.

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$ npm init -y
Wrote to /home/1rv24mc054_khushijs/DEVOPSLAB/program4/package.json:
```

```
{
  "name": "program4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$
```

Step 2: Create [index.js](#) file.



```
Nov 21 10:10 AM
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program4
GNU nano 7.2 index.js
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  res.send("Hello from Git + Docker Integration!");
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});

[ Read 12 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo      M-6 Copy
```

Step 3: Add dependency

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$ npm install express
```

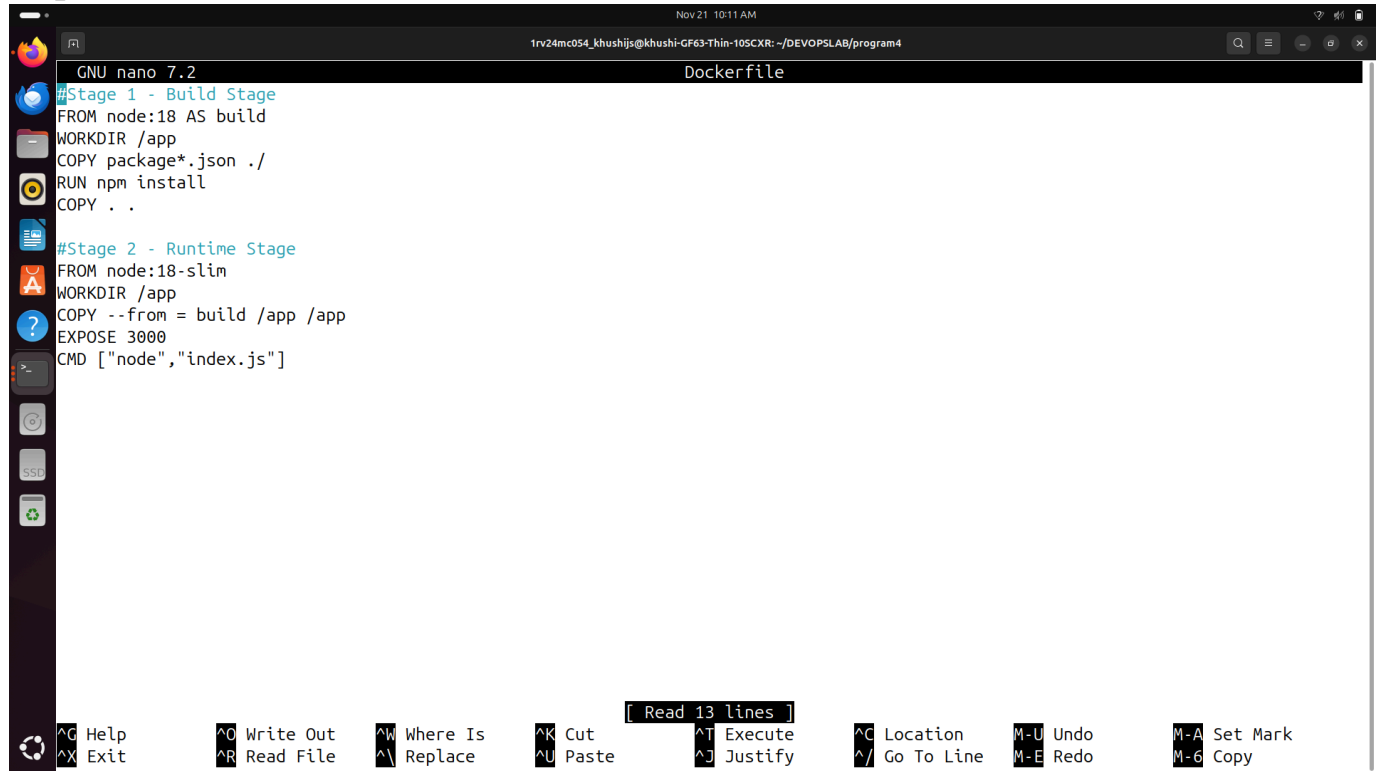
added 68 packages, and audited 69 packages in 4s

18 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$
```

Step 4: Create Dockerfile



```
GNU nano 7.2 Dockerfile
#Stage 1 - Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

#Stage 2 - Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]

[ Read 13 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        [ Read 13 lines ] ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify      ^_ Go To Line  M-E Redo      M-6 Copy
```

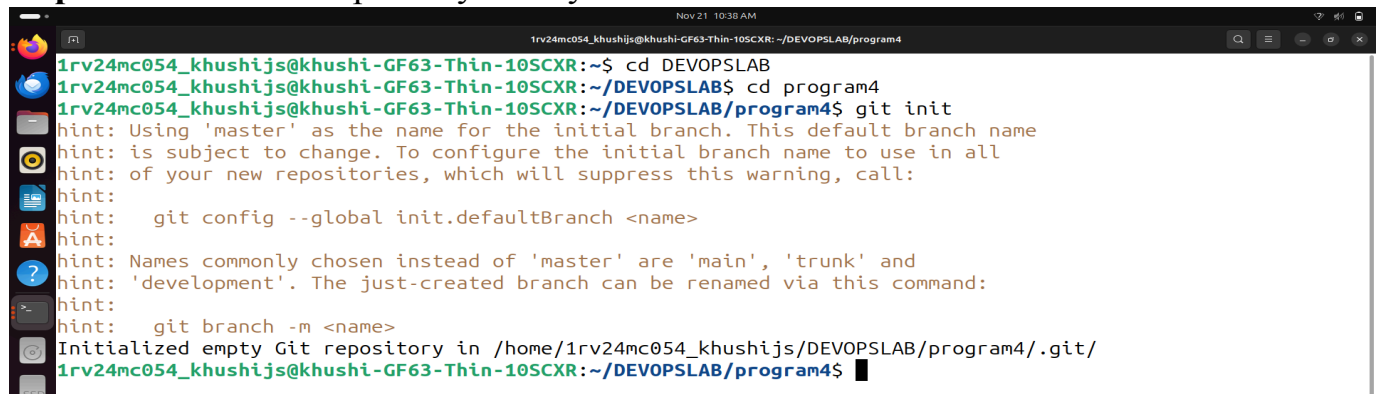
Step 5: Create a new repo on GitHub

Click on create new->new repository->Add a repository name

Step 6: Getting a Personal Access Token(PAT)

Go to profile->Developer Settings->PAT->Tokens(Classic)->Give repo permissions->Generate token

Step 7: Initialize Git repository locally



```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~$ cd DEVOPSLAB
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ cd program4
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /home/1rv24mc054_khushijs/DEVOPSLAB/program4/.git/
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$
```

```
Nov 21 10:41 AM
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program4
agavi"
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$ git commit -m "Initial commit"
[master (root-commit) 6513e76] Initial commit
636 files changed, 72754 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/History.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
```

Step 8: Link to remote repository

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$ git branch -M main
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$ git remote add origin https://github.com/khushijsampagavi/program4.git
```

Step 9: Push code to GitHub

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$ git push -u origin main
Username for 'https://github.com': khushijsampagavi
Password for 'https://khushijsampagavi@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 8 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 724.75 KiB | 4.83 MiB/s, done.
Total 719 (delta 140), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (140/140), done.
To https://github.com/khushijsampagavi/program4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program4$
```

